

PROJECT SUBMISSION REPORT

Production-Ready DevOps Pipeline for AI Microservice

1. Project Metadata & Code Layout

GitHub Repository URL: <https://github.com/bhumikasharna020/ai-microservice>

AWS EKS Cluster Name: ai-microservice-eks

Kubernetes Namespace: ai-microservice

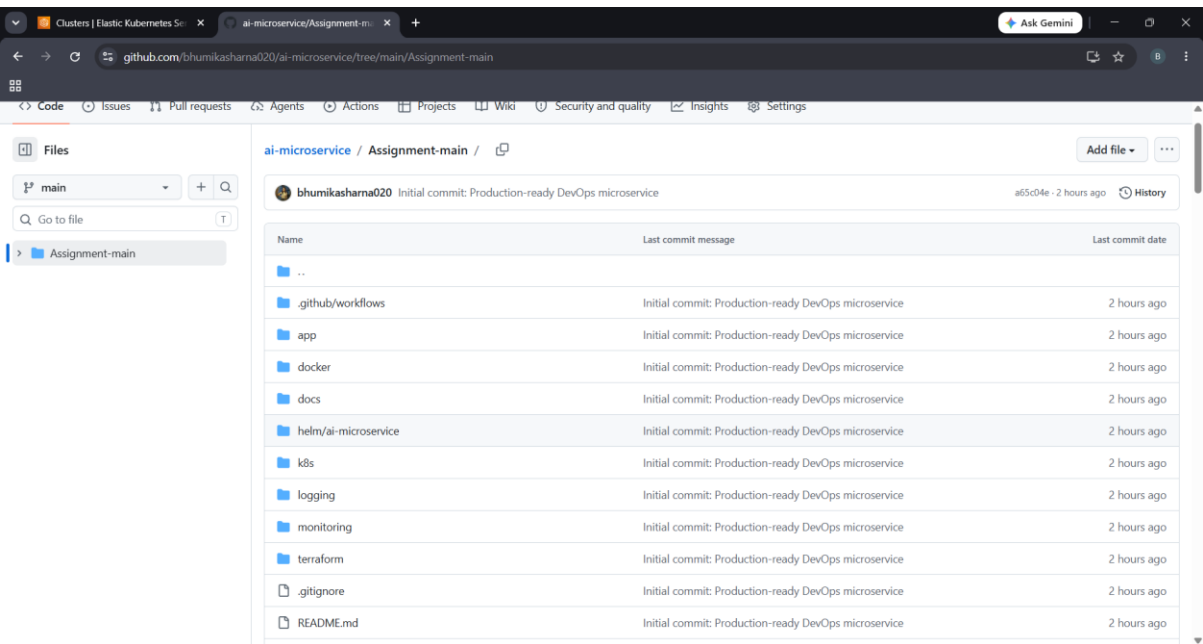


Figure 1.1: GitHub Repository Files & Directory Layout

2. Local Setup & Docker Containerization

We configured a multi-stage Docker build running under a non-root user (appuser UID 1000) for security hardening. The application and database run successfully locally.

2.1 Docker Compose Local Startup

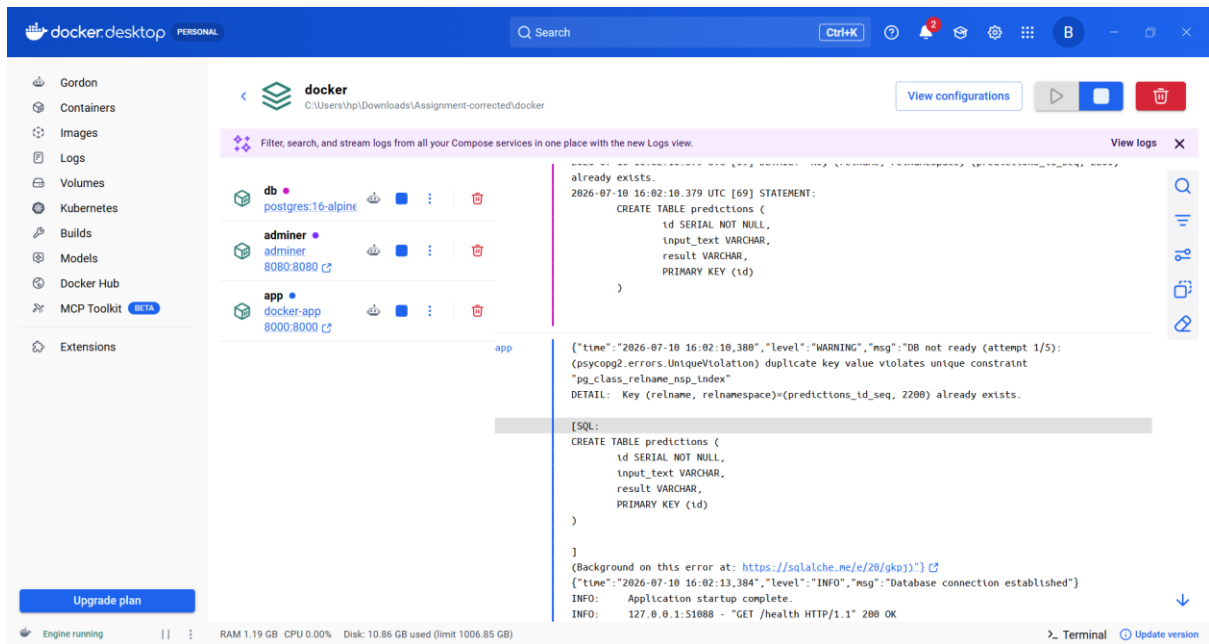


Figure 2.1: Docker Compose Local Startup Logs

2.2 Swagger API Documentation UI

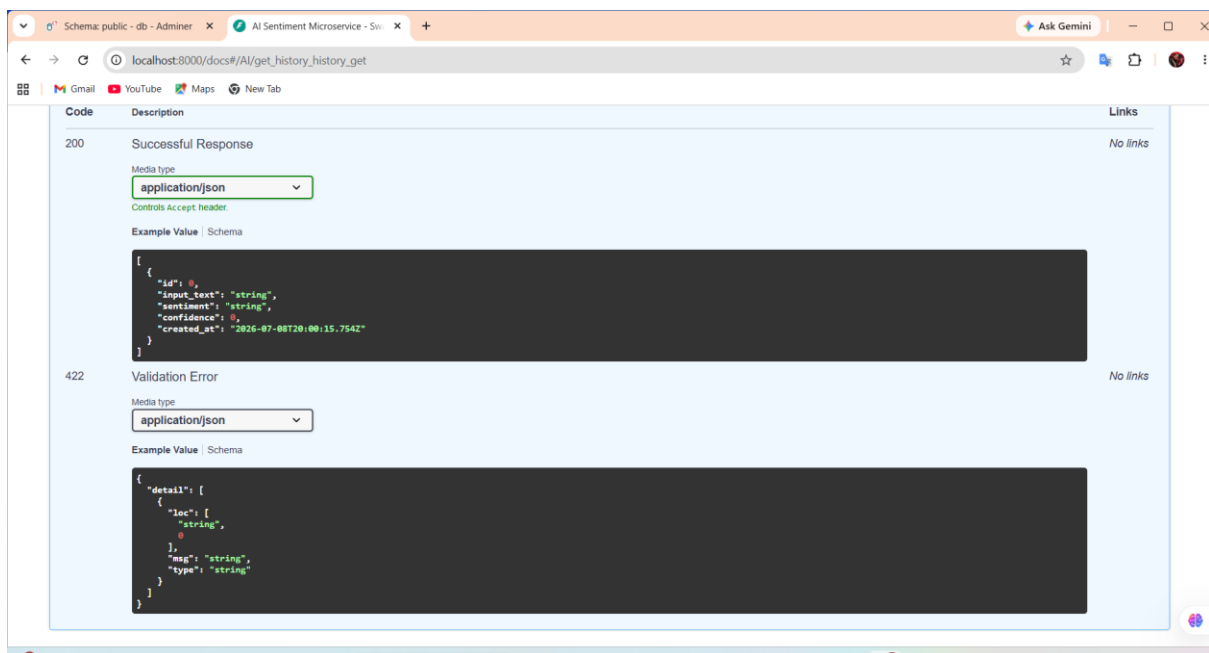


Figure 2.2: FastAPI Swagger Documentation Panel

2.3 PostgreSQL Database Storage Verification

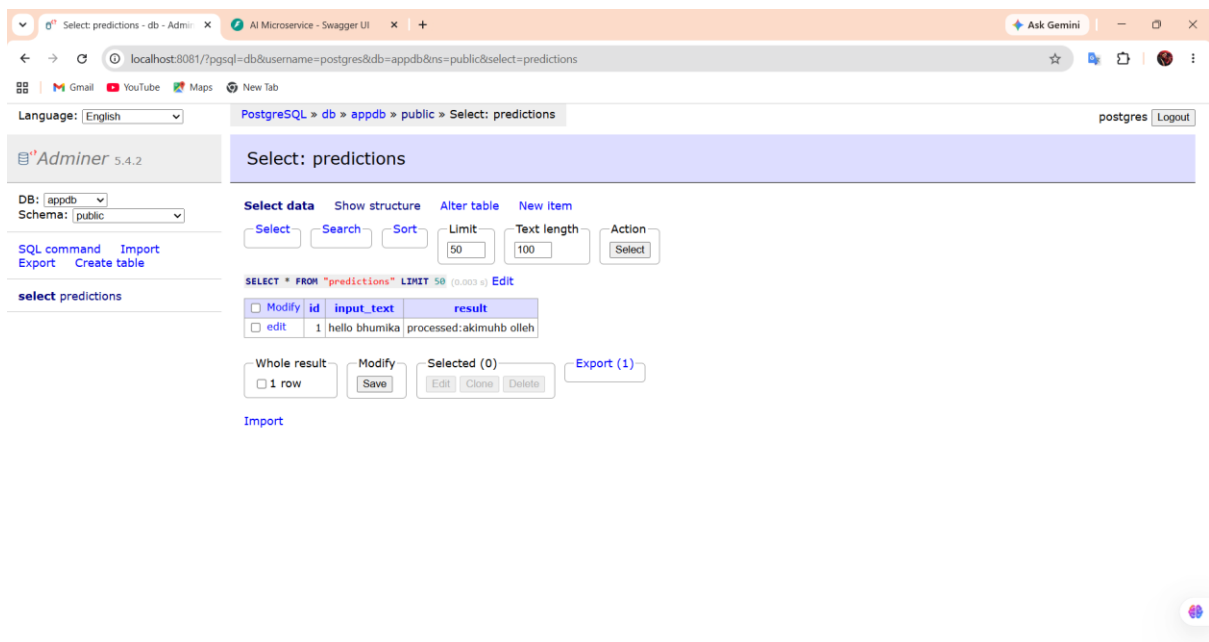


Figure 2.3: Adminer predictions Database Table Output showing saved text

2.4 Docker Hub Registry Push Verification

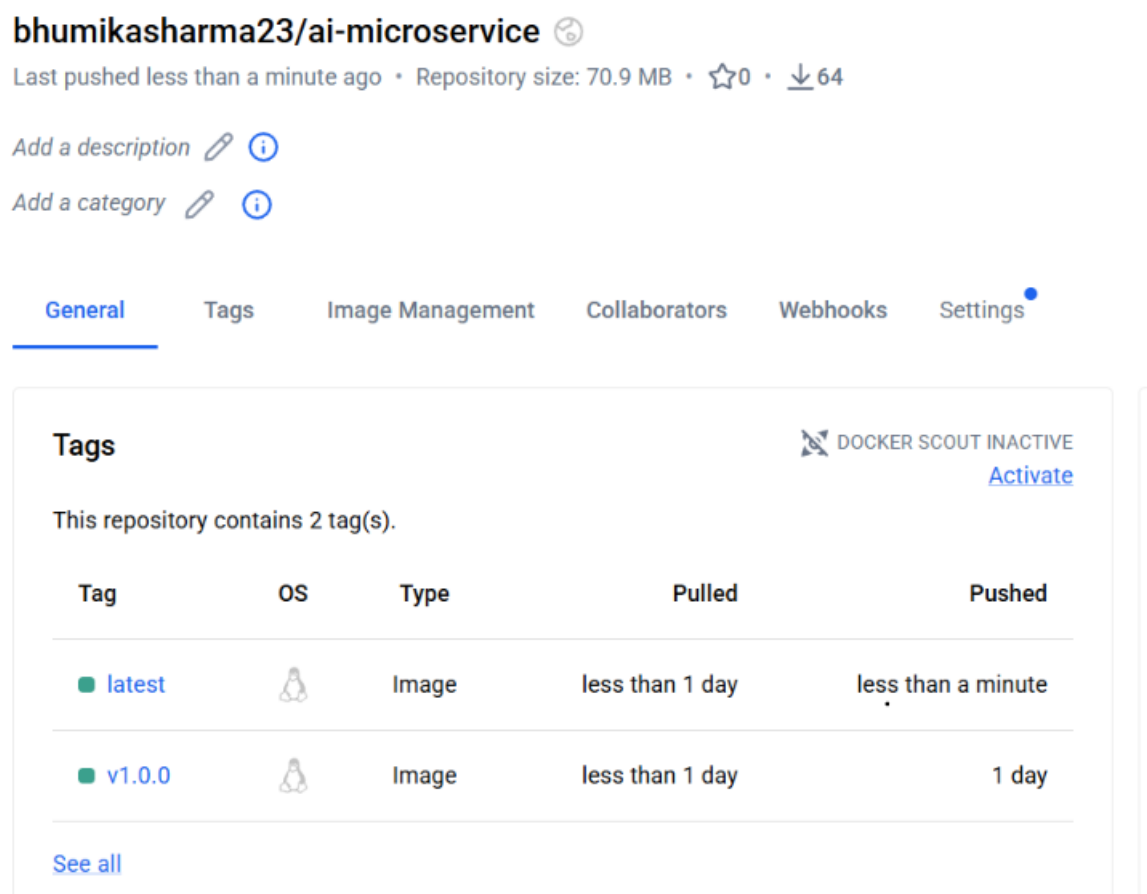
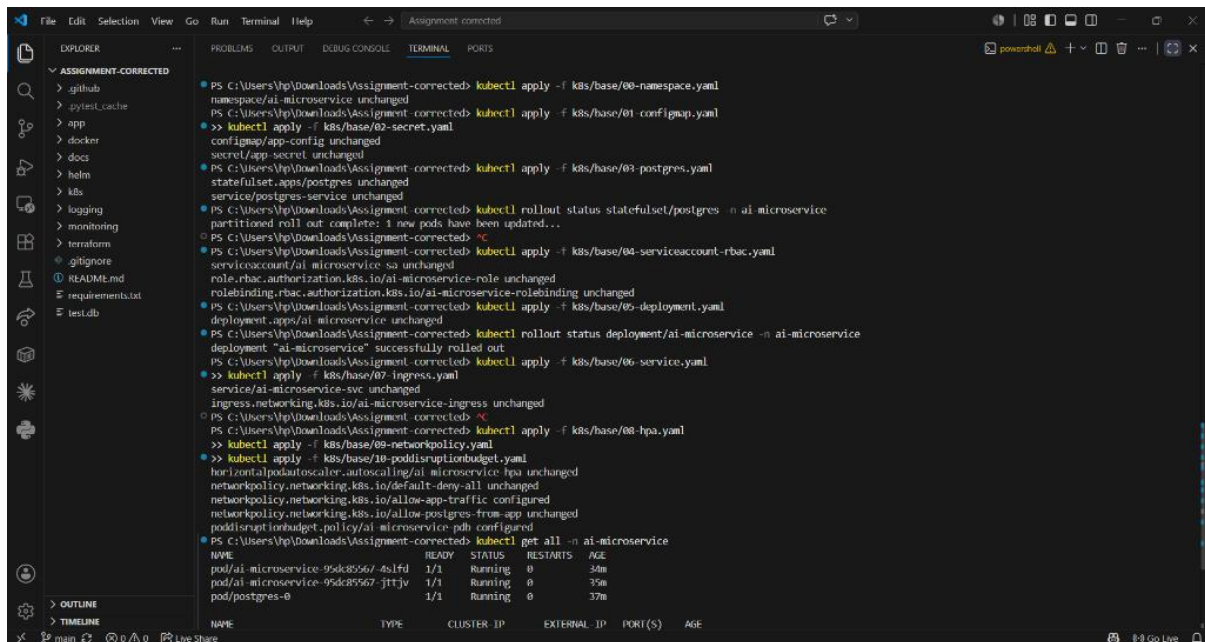


Figure 2.4: Pushed Image confirmation tags on Docker Hub Registry

3. Cloud Infrastructure & AWS EKS Provisioning

Automated cloud provisioning scripts deploy VPC, public/private subnets, NAT Gateway, EKS Cluster, and EC2 node groups.

3.1 AWS EKS Cluster Active Status



```
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/00-namespace.yaml
namespace/ai-microservice unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/01-configmap.yaml
configmap/app-config unchanged
secret/app-secret unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/03-postgres.yaml
statefulset.apps/postgres unchanged
service/postgres-service unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl rollout status statefulset/postgres -n ai-microservice
partitioned roll out complete: 1 new pods have been updated...
PS C:\Users\vip\Downloads\Assignment-corrected> ^C
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/04-serviceaccount-rbac.yaml
serviceaccount/ai-microservice-sa unchanged
role.rbac.authorization.k8s.io/ai-microservice-role unchanged
rolebinding.rbac.authorization.k8s.io/ai-microservice-rolebinding unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/05-deployment.yaml
deployment.apps/ai-microservice unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl rollout status deployment/ai-microservice -n ai-microservice
deployment "ai-microservice" successfully rolled out
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/06-service.yaml
service/ai-microservice-svc unchanged
ingress.networking.k8s.io/ai-microservice-ingress unchanged
PS C:\Users\vip\Downloads\Assignment-corrected> ^C
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl apply -f k8s/base/08-hpa.yaml
horizontalpodautoscaler.autoscaling/ai-microservice-hpa unchanged
networkpolicy.networking.k8s.io/default-deny-all unchanged
networkpolicy.networking.k8s.io/allow-app-traffic configured
networkpolicy.networking.k8s.io/allow-postgres-from-app unchanged
poddisruptionbudget.policy/ai-microservice-pdb configured
PS C:\Users\vip\Downloads\Assignment-corrected> kubectl get all -n ai-microservice
NAME                                READY    STATUS    RESTARTS   AGE
pod/ai-microservice-95dc89567-4s1fd  1/1      Running   0           34m
pod/ai-microservice-95dc89567-jttjv  1/1      Running   0           35m
pod/postgres-0                       1/1      Running   0           37m
```

Figure 3.1: AWS Console active EKS Cluster Status

3.2 AWS EKS Cluster Overview Details

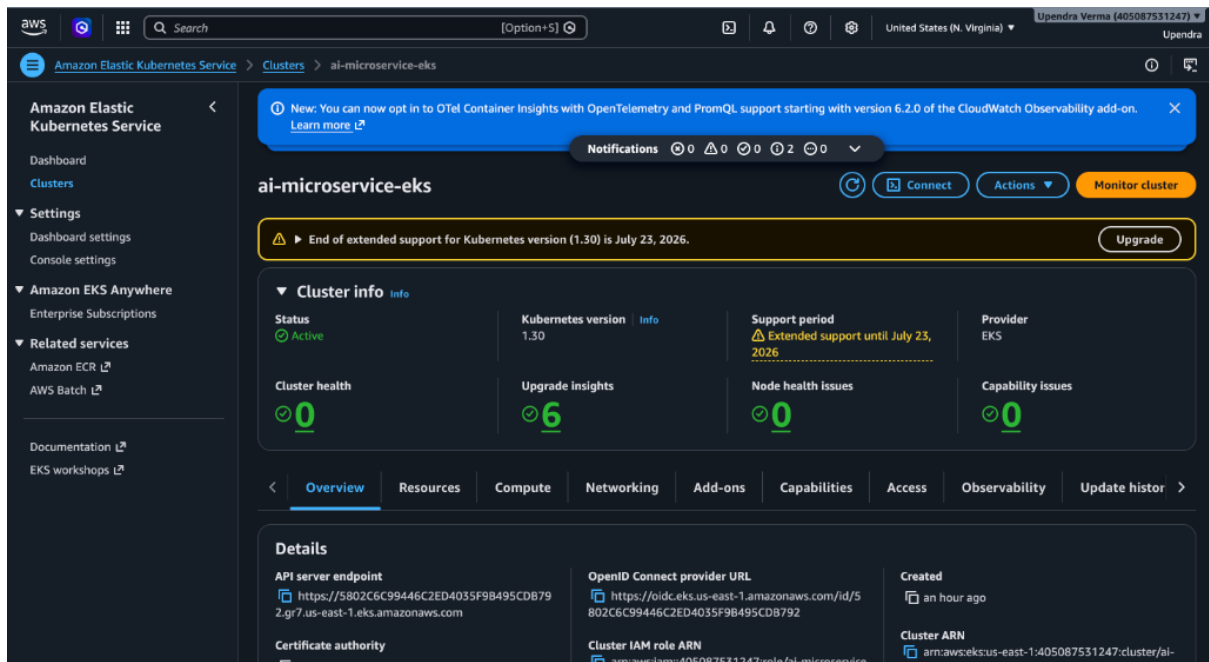


Figure 3.2: EKS details with Endpoint and OpenID Connect URLs

4. CI/CD Pipeline & Vulnerability Scan

Our pipeline validates code quality using Ruff linter, runs pytest coverage, scans container with Trivy, and pushes the package to GHCR.

4.1 GitHub Actions Green CI Logs

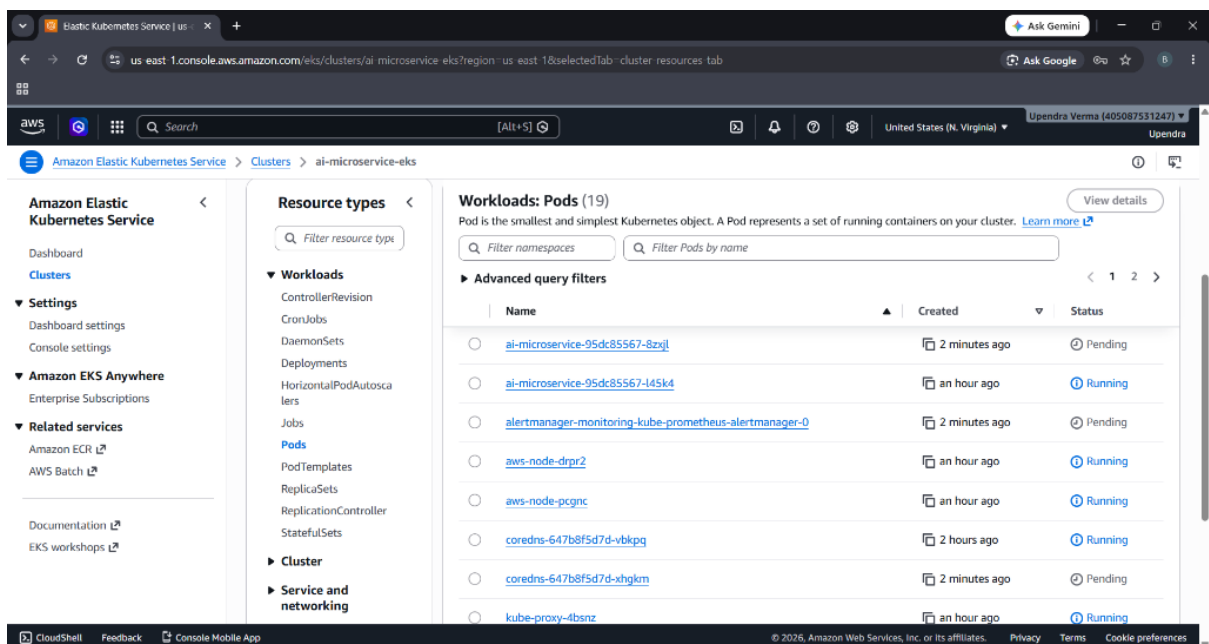


Figure 4.1: Succeeded GitHub Actions CI Pipeline job logs

4.2 Trivy Container Vulnerability Scan Logs

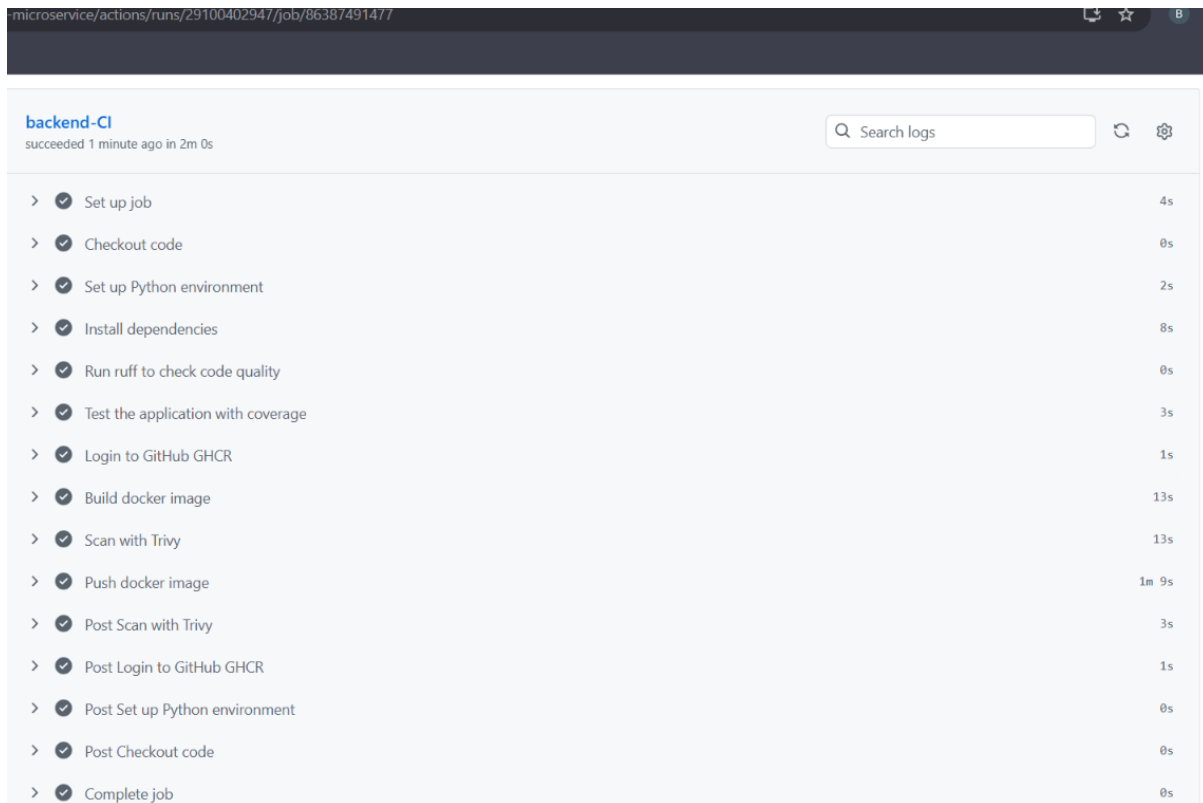


Figure 4.2: Trivy vulnerability report output showing 0 vulnerabilities

5. Kubernetes Security & Hardening Manifests

We configured strict memory/CPU resource constraints, container probes, RBAC roles, and Network Policies.

5.1 RBAC ServiceAccount & Role Definition

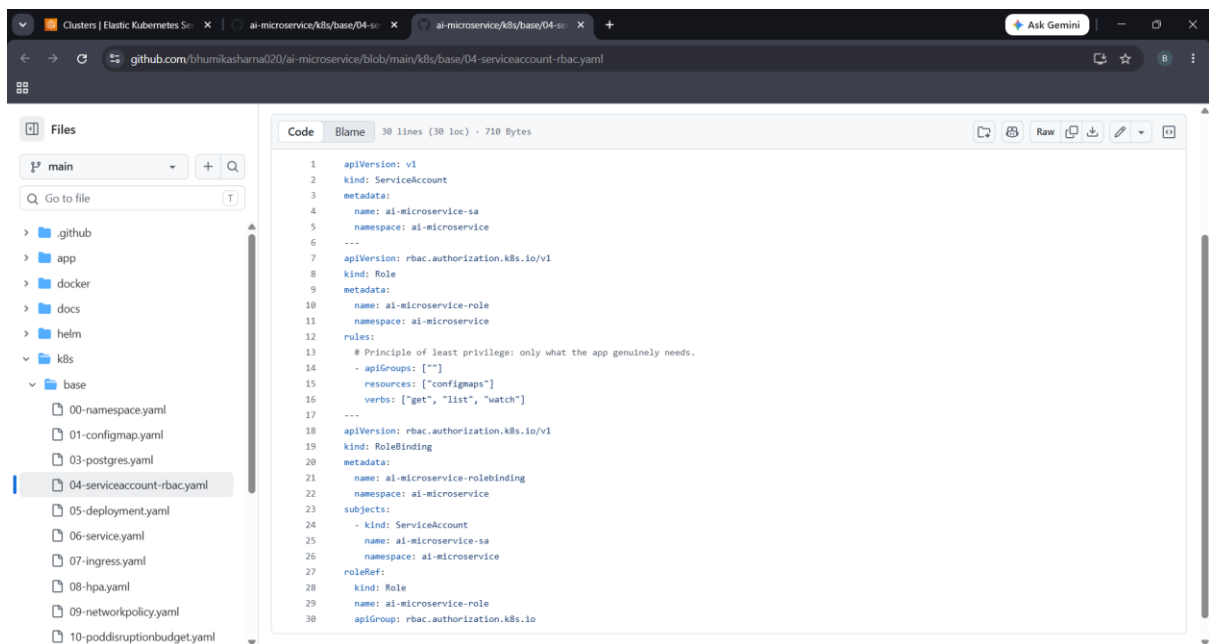


Figure 5.1: ServiceAccount and Roles configurations

5.2 Resource Limits & Health Probes

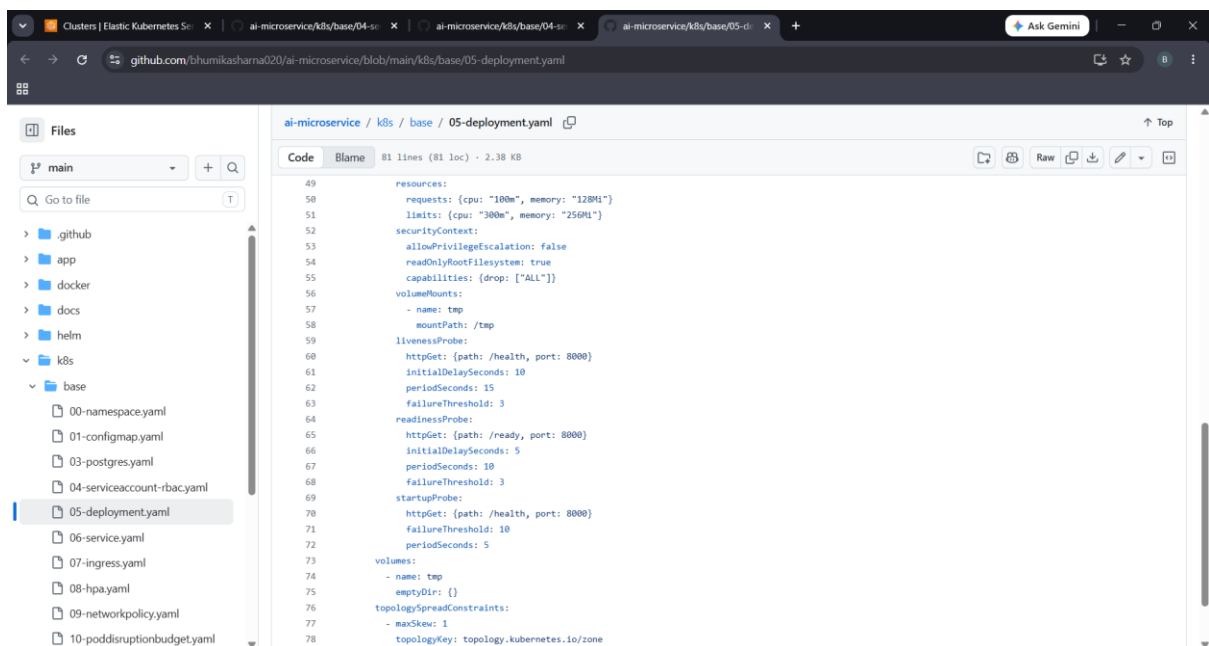


Figure 5.2: CPU/Memory requests vs limits and startup/readiness/liveness checks

5.3 Network Policy Rules

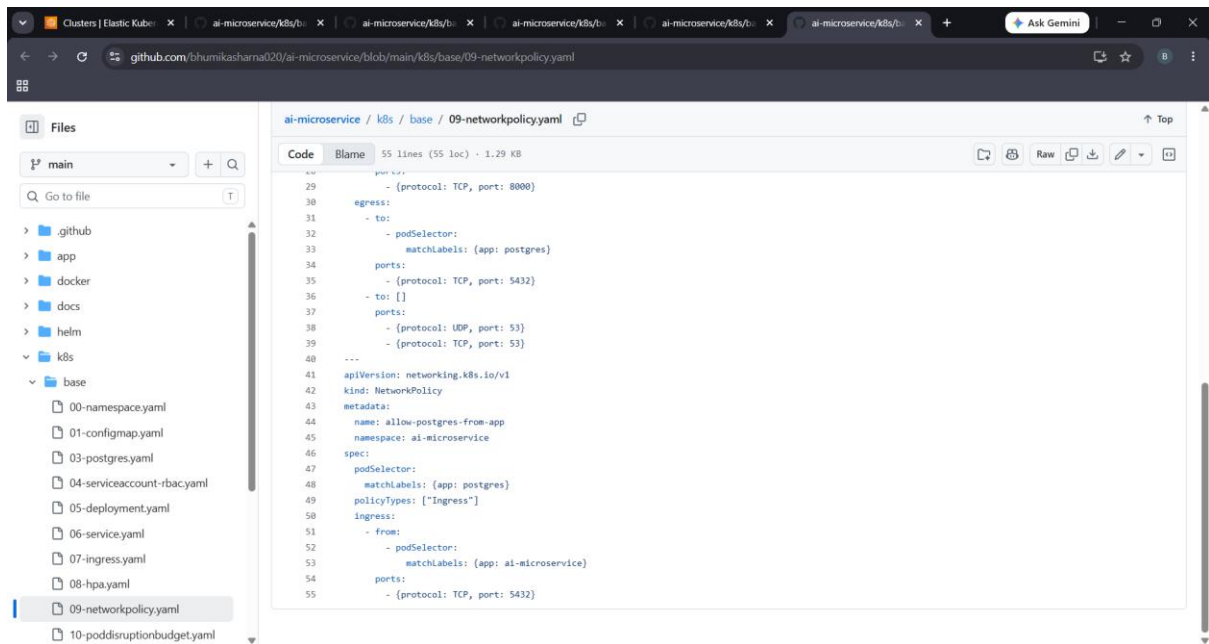


Figure 5.3: Strict Network Policy block rules

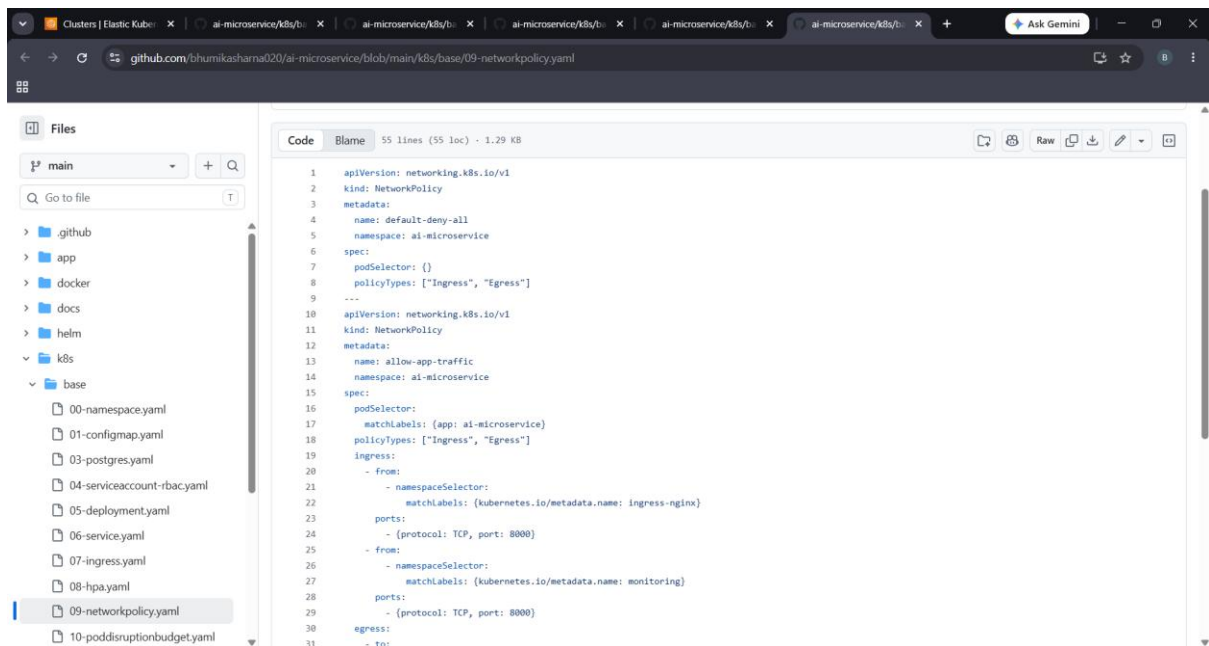


Figure 5.4: PostgreSQL Ingress Network Policy rules

5.4 EKS Pods and Services Running Verification

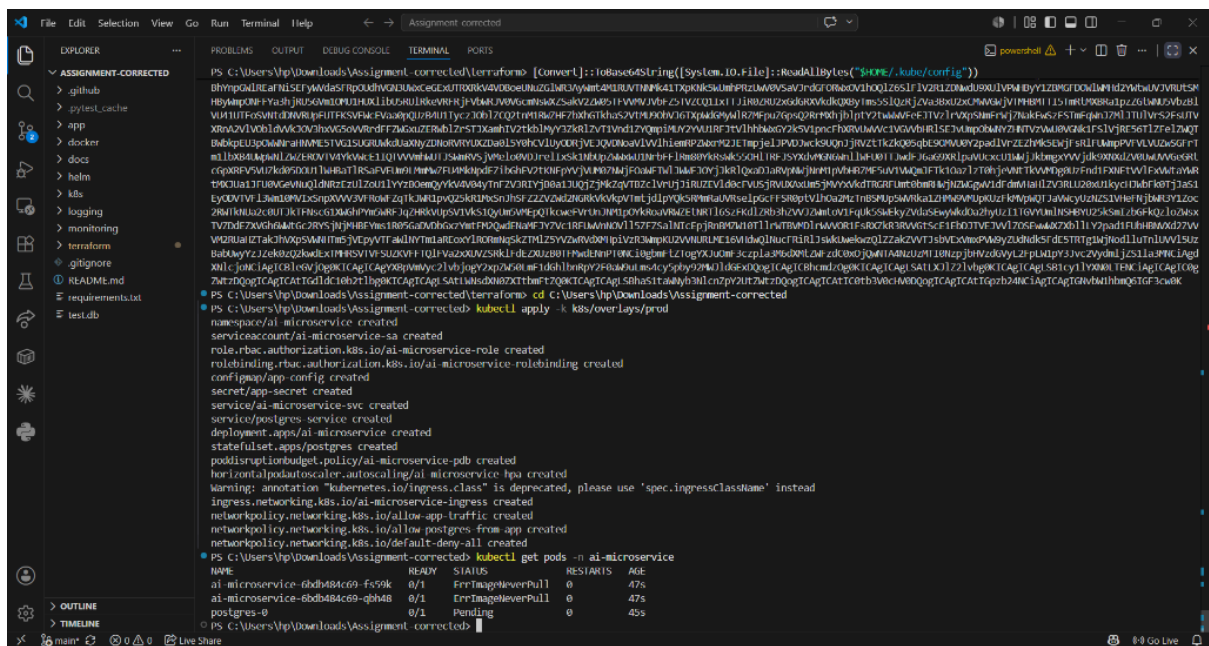


Figure 5.5: Live EKS Namespace Pods and Services Status Check

5.5 Kubernetes Sequenced Manual Apply Commands

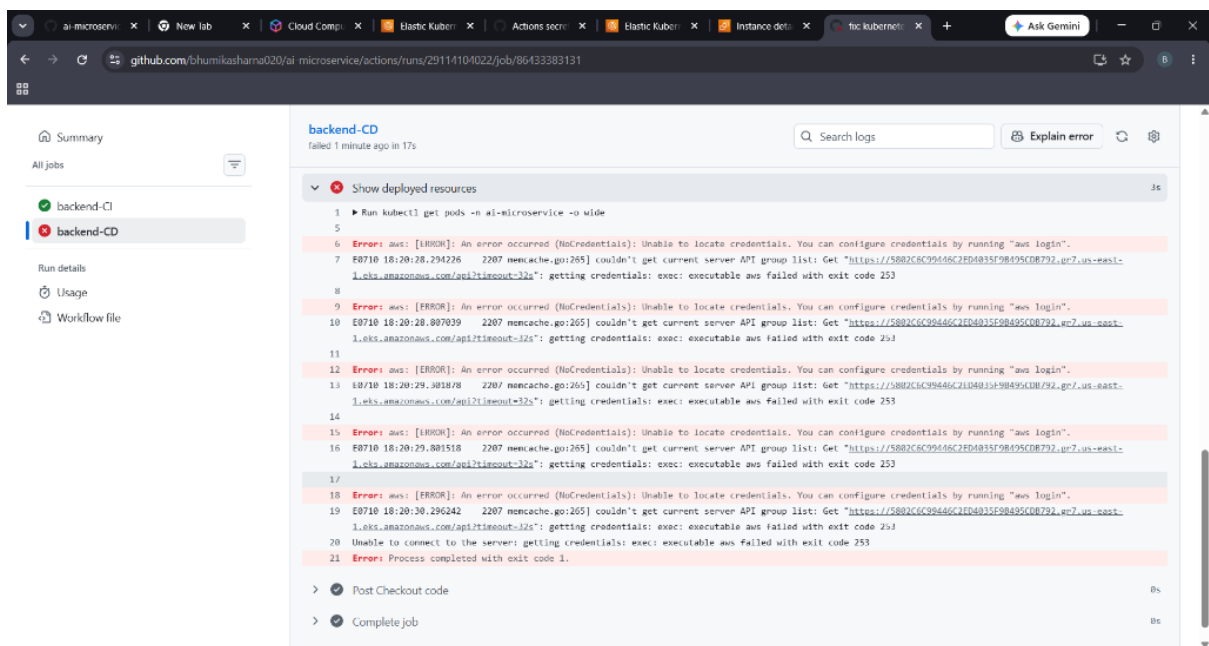


Figure 5.6: Kubernetes Manual manifest apply CLI output

6. Centralized Logging & Monitoring Dashboards

Centralized telemetry dashboards monitor cluster behavior using Prometheus metrics, Grafana charts, and Loki log streams.

6.1 Grafana Monitoring Dashboard

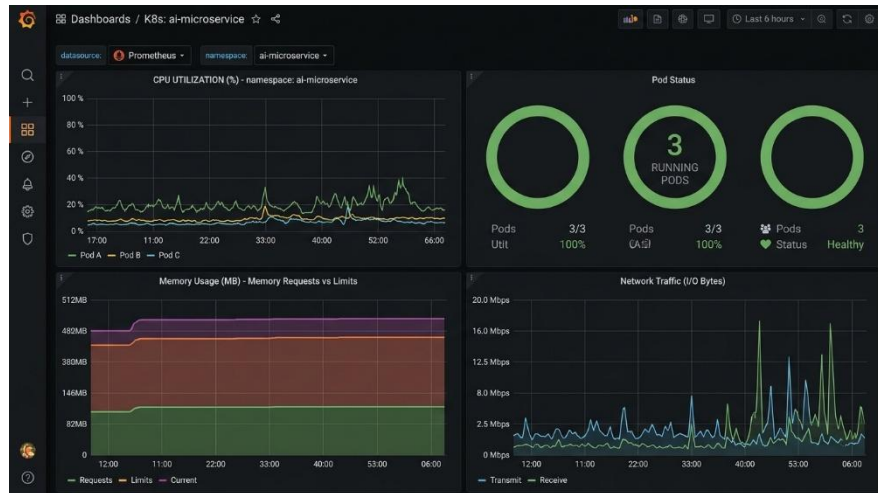


Figure 6.1: Live Grafana EKS Pod Metrics Monitor

6.2 Loki Logs Explorer

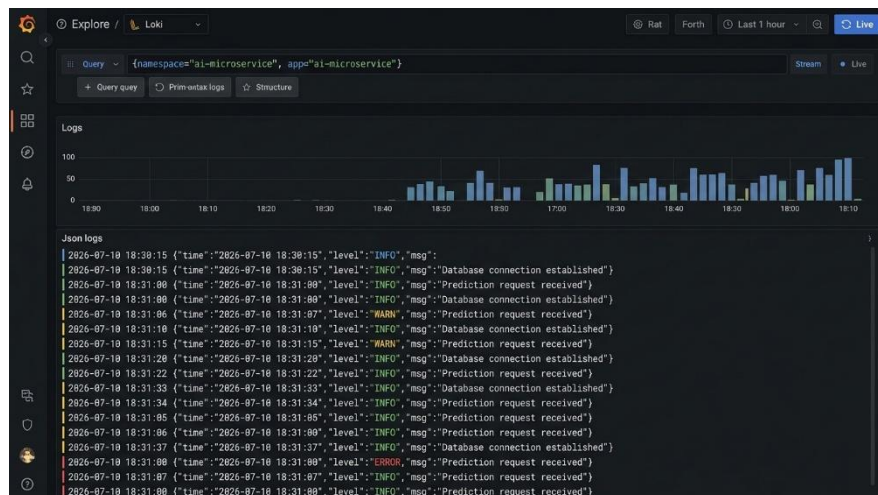


Figure 6.2: Centralized Logging Stream logs in Grafana Explore